AssistAnt Orchestrator Routing Technical Summary

Angeles University Foundation · PTF60 Project · BS Computer Science 4

Request	Write a technical summary report on how AssistAnt's Orchestrator routes work to its agents and models, including the fallback chain.
Skill	Technical report
Routing	hard task, best tier
Models	Kilo (keyless), plain rules
Review	2 round(s), approved, 42/42 claims grounded
Generated	2026-11-05 23:02

Summary

AssistAnt is a personal agent that wraps up the day in Notion and plans tomorrow, running as a Next.js website and an installable app from one project [S1]. Every request goes to the Orchestrator, which matches the request against routines and either runs a matched routine's flow or plans the request itself with the unplanned flow [S1]. The Orchestrator hands work to eleven subagents, each assigned a tier that decides which line of models it asks for [S1]. Each agent asks the router for a tier (fast, smart, or best), and the router walks that tier's chain of providers, using the first one that is available [S8]. If every provider in a chain fails or is unavailable, the router returns nothing and the agent falls back to plain rules, keeping AssistAnt working offline and for free [S10].

Key findings

- The Orchestrator routes work to eleven subagents, of which seven are on the code tier and never call a model, so they are instant and free [S1], [S6].
- The eleven agents are: Reader, Summarizer, Matcher, Notes Scout, Chore Keeper, Planner, Writer, Checker, Grocery Clerk, Briefer, and Portfolio Clerk [S6].
- Tier assignments are: fast tier — Grocery Clerk; smart tier — Summarizer and Briefer; best tier — Notes Scout; code tier — Reader, Matcher, Chore Keeper, Planner, Writer, Checker, and Portfolio Clerk [S6].
- The best chain is Claude Code on the user's Pro or Max plan, Groq gpt-oss-120b, the free providers from the awesome-free-llm-apis list, OpenRouter, then a local Ollama model [S8].
- The smart chain is Gemini Flash, Groq gpt-oss-120b, the free list, OpenRouter, Ollama, then the paid Claude API, which is off by default [S8].
- The fast chain is Groq gpt-oss-20b, Gemini Flash-Lite, Groq Qwen, the free list, OpenRouter, Ollama, then the Claude API [S8].
- Before calling a provider, the router checks its health; only providers reported as ok are called [S8].
- Health statuses are: ok, full when used up today, off when switched off in Sources, nokey when no API key is set, or offline when Ollama or Claude Code is not running [S8].
- Usage is counted per provider per day and resets at midnight in the configured time zone [S10].
- Providers with a daily cap, such as Gemini Flash at 20 requests a day or Groq at 1,000, are skipped once the cap is reached [S10].
- A 429 response marks the provider exhausted for the rest of the day [S10].

- Three keyless gateways from the free list — Kilo Code, LLM7.io and OVHcloud AI Endpoints — work without an API key but are off by default because the list says they may log prompts [S10].
- When no routine matches, the Orchestrator answers the request itself and drafts a reusable routine, asking for approval before saving it [S2].
- Routines are markdown files in the routines folder, each starting with a front matter block containing a name, a flow, and a list of triggers that are case-insensitive regular expressions [S5].
- The Orchestrator tests every routine's triggers against the request and runs the first routine that matches [S5].

How it works

- A user request arrives at the Orchestrator [S1].
- The Orchestrator tests the request against every routine's triggers, which are case-insensitive regular expressions, and runs the first routine whose triggers match [S5].
- If a routine matches, its flow runs, handing work to the subagents named in the flow [S1].
- If no routine matches, the Orchestrator plans the request itself with the unplanned flow and later offers to save a new routine, asking for approval before saving it [S1], [S2].
- Each subagent that needs a model asks the router for a tier: fast, smart, or best [S8].
- The router walks that tier's chain of providers in order and uses the first one that is available [S8].
- Before calling a provider, the router checks its health and only proceeds if the status is ok [S8].
- If a provider has reached its daily cap or returned a 429, it is skipped or marked exhausted for the rest of the day [S10].
- If every provider in the chain fails or is unavailable, the router returns nothing and the agent falls back to plain rules [S10].
- Each request is a run, and a run emits events such as the plan, messages between agents, work done, and the result, which the browser streams live with Server-Sent Events [S9].

Risks and limitations

- Agents on the code tier never call a model, so they are instant and free but cannot perform language-based tasks [S1].
- The paid Claude API in the smart chain is off by default, so it is not used unless explicitly enabled [S8].
- The keyless gateways Kilo Code, LLM7.io and OVHcloud AI Endpoints are off by default because the free list says they may log prompts [S10].
- Providers with daily caps, such as Gemini Flash at 20 requests a day and Groq at 1,000, are skipped once reached, which can force fallback to slower or less capable providers [S10].
- A 429 response marks a provider exhausted for the rest of the day, so a transient error can remove a provider from the chain until midnight [S10].
- If every provider in a chain fails, the agent falls back to plain rules, which may produce lower-quality results [S10].

- In public demo mode (PUBLIC_DEMO=1), only the keyless model gateways are offered, switched on, and no API key is ever sent; when they are busy, flows fall back to plain rules [S7].
- The public demo runs on the free tier of Render [S7].
- Scheduled jobs (finish day at 22:10 and morning brief at 07:00) run only in the configured time zone; the scheduler checks the clock every 20 seconds and runs each job once per day [S2].
- Risky changes wait for user approval: the Writer holds new guessed Reminders and the deletion of today's block until the user approves them, and scheduled runs keep their approvals waiting until the app is opened [S9].
- The Notion workspace retries on 429 responses with a growing delay, but repeated failures could block flows [S3].
- The demo workspace is a fictional user's Tasks page with sample tasks, reminders, and notes, seeded once per day in memory [S3], [S7].

Sources

- [S1] architecture.md, part 1
- [S2] routines.md, part 3
- [S3] workspace.md, part 1
- [S5] routines.md, part 1
- [S6] architecture.md, part 2
- [S7] public-demo.md, part 1
- [S8] model-router.md, part 1
- [S9] architecture.md, part 3
- [S10] model-router.md, part 2

How the team worked

You → **Orchestrator** · New request
From now on, always end reports with a 'Risks and limitations' section. Write a technical summary report on how AssistAnt's Orchestrator routes work to its agents and models, including the fallback chain.

Orchestrator → **Memory** · Save standing rule
From now on, always end reports with a 'Risks and limitations' section.

Orchestrator → **Reader** · Assign the crew
Reader Notes Scout Writer Checker Publisher

Reader → **Notes Scout** · Notes to verify
20 notes from 5 files

Notes Scout → **Writer** · Evidence pack
10 passages, S1–S10

Writer → **Checker** · Draft for review
round 1

Checker → **Writer** · Send back for revision
Not supported by the evidence (best match 0.47), remove or rephrase: "The public demo runs on the free tier of Render, which may impose perf…"

Writer → **Checker** · Draft for review
round 2

Checker → **Publisher** · Approved
approved

Publisher → **You** · Report ready
1117 words, 9 sources